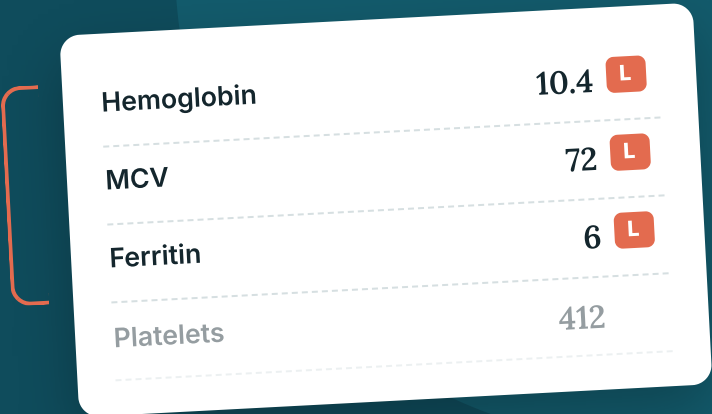




Hemoglobin	10.4	L
MCV	72	L
Ferritin	6	L
Platelets	412	

LAB REPORTS DECODED

A Practical Guide to Understanding Blood, Urine & Common Laboratory Reports

Learn the numbers. Recognize the patterns. Understand the report.

MEDICAL REFERENCE SERIES

Lab Reports Decoded

A Practical Guide to Understanding Blood, Urine &
Common Laboratory Reports

From Individual Values to Complete Report Patterns

2026 Edition · Medical Reference Series

Lab Reports Decoded

A Practical Guide to Understanding Blood, Urine & Common Laboratory Reports

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All sample laboratory reports in this book are fictional and were created for teaching. Any resemblance to real patients is coincidental.

MEDICAL DISCLAIMER

This book is for education and general understanding of laboratory reports. It does not replace consultation, diagnosis, treatment or individualized advice from a qualified healthcare professional.

Laboratory values must never be interpreted in isolation. Results are influenced by age, sex, pregnancy, medications, hydration, diet, exercise, timing of collection, acute illness, laboratory method and other factors. Reference intervals differ between laboratories: always use the interval printed on the actual report.

An abnormal result does not automatically establish a disease, and a result within a reference interval does not necessarily exclude one. Do not start, stop or change any medication based on this book.

Diabetes categories follow the American Diabetes Association Standards of Care; kidney categories follow KDIGO. Clinical guidance changes over time — check current recommendations.

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A Note to the Reader

A laboratory report can look deceptively simple. There is a test name. There is a number. There is a unit. And beside it sits a reference interval.

When a number lands outside that interval, the natural reaction is: *"What does this mean?"*

But laboratory interpretation is rarely as simple as comparing one number with another. A slightly abnormal result may be temporary or clinically insignificant. A value inside the reference interval may still deserve attention in a particular situation. And two individually mild abnormalities can become far more meaningful when they appear together.

That is why this book is built around one central idea:



Don't just read the number.
Understand the pattern.

You will learn what common laboratory tests measure, typical adult reference intervals where they are meaningful, why values rise or fall, which tests belong together, and how groups of results form recognizable patterns.

The purpose is not to turn laboratory values into self-diagnoses. The purpose is to make laboratory reports easier to understand — so that conversations with your doctor, your patients or your colleagues become clearer and better informed.

How This Book Is Different

Most lab references are dictionaries: one test, one range, a list of causes, repeat. This book is organized as a course. Each system chapter answers four questions:

01**What are we measuring?**

The biology behind the test.

02**What do the numbers tell us?**

Typical intervals and what higher or lower results can mean.

03**What changes the result?**

Medicines, hydration, fasting, timing, illness and the sample itself.

04**How do the results connect?**

Related markers combined into meaningful patterns.

That fourth step is the heart of the book. A single low hemoglobin tells us less than this combination:



The second picture points in a much clearer direction.

Recurring features

DECODE THIS

A card for one marker: what it measures, its typical interval, what pushes it up or down, and what to read alongside it.

PATTERN LAB

A combination of results and the direction it points — plus the question to ask next.

REPORT ROOM

A fictional report you read step by step: first look, second look, third look, then interpretation.

**RANGE DEPENDS ON
CONTEXT**

A badge for tests where one universal number would mislead. Use the laboratory's own interval.

Important Medical Note

This book is an educational reference. It does not provide medical diagnosis or individualized treatment.

Reference intervals vary between laboratories because of differences in analytical methods, equipment and reference populations. Age, sex, pregnancy and clinical circumstances also change expected values.

HOW TO USE THE NUMBERS IN THIS BOOK

Whenever this book gives a

TYPICAL ADULT INTERVAL

, use it for learning — never as a replacement for the interval printed on an actual laboratory report. Values marked with “~” are approximate.

A laboratory result should always be interpreted together with symptoms, history, medications, physical examination, previous results and other investigations. If a result worries you, the right next step is a conversation with a qualified healthcare professional — not a search engine, and not this book alone.

THE 5-QUESTION METHOD

- 1 WHAT?**
What does this test actually measure?
- 2 WHERE?**
Which organ, cell type or system does it represent?
- 3 WHY?**
Why could it be higher or lower than expected?
- 4 WITH WHAT?**
Which other results should be read alongside it?
- 5 WHAT NEXT?**
Does it call for correlation, a repeat test, another investigation or clinical review?

You will meet the 5-Question Method throughout the book. Whenever a value is unfamiliar, run it through these five questions before drawing any conclusion.

PART I

Before You Read the Numbers

How laboratory reports actually work — and why a flag is only the start of the story.

- Anatomy of a lab report
- What changes a result
- Why “normal” isn't always normal
- The three layers of interpretation

Anatomy of a Lab Report

Before learning a single test, learn to see what the page is actually showing you.

Every quantitative result on a laboratory report is built from the same five parts. Once you can see them clearly, any report — from any laboratory — becomes easier to navigate.

TEST NAME	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Potassium	5.2	mmol/L	3.5 – 5.0	H
① What was measured	② The patient's value	③ The scale	④ This lab's expected range	⑤ Outside the range?

Test name

The parameter being measured. The same test may appear under different names or abbreviations (for example, FBS, FPG and “fasting glucose”).

Result

The value measured in the patient's sample at that moment.

Unit

The measurement system. Glucose in mg/dL and in mmol/L are different numbers for the same concentration — always check units before comparing results.

Reference interval

The range established by *that* laboratory for an appropriate reference population — usually the central 95% of healthy people.

Flag

A marker such as H, L, * or an arrow, showing that the result lies outside the interval.

The most common interpretation mistake

Look at the potassium result above: **5.2 mmol/L, flagged H**. The tempting conclusion is that something is wrong. But before any interpretation, a careful reader asks:

❓ Was the sample hemolyzed? Damaged red cells release potassium into the sample.

❓ What reference interval did this laboratory use?

- ? What is the kidney function?
- ? Which medicines are being taken?
- ? What were the previous potassium values?
- ? Are there symptoms or ECG changes?

KEY IDEA

The flag tells you the result crossed a numerical boundary. It does not tell you *why*.

Reference interval vs decision limit

These two ideas are often confused, and the difference matters.

Reference interval

Describes the values found in a defined, usually healthy, reference population. It answers: *"Is this result typical?"*

Clinical decision limit

A threshold used for diagnosis, risk assessment or treatment decisions. It answers: *"Does this result cross a line that changes management?"*

Diabetes is the clearest example. It is diagnosed using established criteria (such as fasting glucose ≥ 126 mg/dL or HbA1c $\geq 6.5\%$), not simply by whether the glucose falls outside the interval printed on a report. LDL cholesterol goals and troponin cut-offs work the same way.

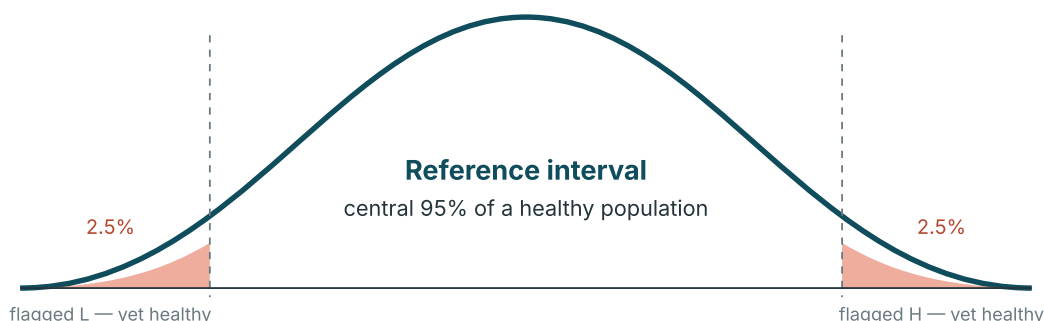
“Normal” Isn't Always Normal

02

A reference interval is a statistical landmark — not a certificate of health.

The word *normal* creates more confusion on lab reports than any other.

Most reference intervals are built by testing a group of healthy people and taking the middle 95% of their results. That has a surprising consequence: by design, about **1 in 20 healthy people** will fall outside the interval for any single test.



Now imagine a 20-test panel. The chance that a perfectly healthy person has at least one flag is far higher than most people expect. That is why a single mild flag, with everything else consistent, is often less important than it looks.

The reverse is also true. A value inside the interval cannot rule out every disease. A creatinine that doubles from 0.6 to 1.2 mg/dL may still sit within many laboratories' intervals — yet it may represent a meaningful change for that person.

KEY IDEA

Think of the reference interval as a statistical and clinical reference point — not an automatic diagnosis line.

Three things that move the line itself

The laboratory

Different analyzers, reagents and populations produce different intervals.

The person

Age, sex, pregnancy, ancestry and altitude shift what is expected.

The purpose

Screening, diagnosis and treatment targets may each use different cut-offs.

Why Lab Results Change

03

Laboratory values are biological measurements — and biology moves.

The same person, tested twice, rarely gets identical results. Some variation is analytical; much of it is biological. Knowing the common influences prevents many false alarms.

Fasting & meals

Glucose and triglycerides rise after eating. Some tests require 8–12 hours without calories.

Time of day

Cortisol, testosterone and serum iron follow daily rhythms. Timing must match the test.

Medication

Many prescription and over-the-counter medicines alter results — directly or through the organ they affect.

Pregnancy

Physiology changes substantially; many intervals (thyroid, hemoglobin, ALP) differ.

Acute illness

Infection, inflammation and physiological stress cause temporary changes in many markers.

Hydration

Dehydration concentrates the blood, lifting hemoglobin, hematocrit, albumin and urea.

Exercise

Strenuous activity can raise CK, AST, and sometimes creatinine and urine protein.

Supplements

High-dose biotin can interfere with certain immunoassays, including some thyroid and cardiac tests.

Age & sex

Hematology, hormones, creatinine and ALP all vary across populations.

The sample

Hemolysis, delayed processing or the wrong tube can change a result before analysis even begins (Chapter 28).

PRACTICAL TIP

When a result is unexpected and doesn't fit the clinical picture, “repeat the test under proper conditions” is often the most sensible next step.

Why the trend often beats the single value

Because results move, a series tells you more than a snapshot. Always ask: **previous result → current result → rate of change**. A result still within the interval can matter if it has changed substantially from the person's own baseline. Conversely, a small

abnormality that has been stable for years carries a different meaning from one that appeared last week.

The Three Layers of Interpretation

The thinking model you will use in every chapter that follows.

Experienced clinicians rarely read a lab result as a single number. Without always naming it, they move through three layers. This book makes that process explicit.

LEVEL 1

The Number

Is it high, low or within the laboratory's interval? Are the units right? Is the flag real?

LEVEL 2

The Context

Could age, sex, pregnancy, medicines, fasting, hydration, timing or the sample explain it?

LEVEL 3

The Pattern

What do the related results show? Do they point the same way, or tell a different story?

NUMBER**CONTEXT****PATTERN**

A quick worked example

Level 1 — Number. Urea is 52 mg/dL, flagged high.

Level 2 — Context. The person has had vomiting for two days and drank little. Dehydration can raise urea.

Level 3 — Pattern. Hematocrit is at the upper end, urine specific gravity is high, creatinine is only mildly raised. The combination fits concentration from fluid loss — which should be confirmed clinically, and the tests repeated after rehydration.

Notice what happened: the same urea value means something different once the context and the pattern are added. Keep this three-layer habit in mind as we move system by system through the body.

PART II

Reading the Body, System by System

Blood counts, sugar, kidneys, liver, lipids, thyroid, electrolytes and nutrition — each read as a connected panel.

- CBC — your blood cell dashboard
- Iron studies
- Blood sugar
- Kidney panel
- Liver panel
- Lipid profile
- Thyroid tests
- Electrolytes
- Vitamins & nutrition

CBC — Your Blood Cell Dashboard

One request, three cell systems, a dozen numbers. Start by asking which system is off.

A Complete Blood Count is not really one test. It is a collection of measurements describing the three cell families circulating in blood.

RED CELLS

Oxygen transport

Hb · RBC · HCT · MCV · MCH ·
MCHC · RDW

WHITE CELLS

Immune defense

WBC · differential · absolute
counts

PLATELETS

Hemostasis & clotting

Platelet count · indices

START HERE

Don't memorize a CBC line by line. First ask: **which of the three systems is abnormal?** Then read only that system closely.

CBC at a glance

Marker	Typical adult interval*	The question it answers
Hemoglobin	Men ~13–17 g/dL Women ~12–15 g/dL	Is anemia or erythrocytosis present?
RBC count	Men ~4.5–5.9 Women ~4.1–5.1 million/ μ L	How many red cells?
Hematocrit	Men ~40–50% Women ~36–44%	What share of blood volume is red cells?
MCV	~80–100 fL	How big is the average red cell?
MCH	~27–33 pg	How much Hb is in each red cell?
MCHC	~32–36 g/dL	How concentrated is Hb inside the cells?
RDW	~11.5–14.5%	How varied are red cell sizes?

Marker	Typical adult interval*	The question it answers
WBC	~4,000–11,000/ μ L	What is the overall white-cell count?
Platelets	~150,000–450,000/ μ L	How many platelets?

*Approximate educational values. Laboratory-specific intervals take priority.

The red-cell system

DECODE THIS

Hemoglobin Hb

WHAT IT MEASURES The oxygen-carrying protein inside red blood cells. The headline number for anemia.	TYPICAL ADULT INTERVAL Men ~13–17 g/dL Women ~12–15 g/dL
▲ WHEN IT RUNS HIGHER Dehydration (concentration), chronic hypoxia, smoking, high altitude, and true increases in red-cell mass such as polycythemia.	▼ WHEN IT RUNS LOWER Iron, B12 or folate deficiency; blood loss; chronic disease or inflammation; kidney disease; hemolysis; bone-marrow disorders.

READ IT ALONGSIDE MCV → RDW → reticulocytes → ferritin/iron studies → B12/folate

PEARL *Low Hb tells you anemia is present or needs consideration. It doesn't tell you why — the indices and follow-up tests do that.*

RBC count
M ~4.5–5.9 · F ~4.1–5.1 million/ μ L
 Low with anemia, blood loss, marrow disorders or deficiencies. High with dehydration, chronic hypoxia or erythrocytosis. A high count with low MCV can be seen in thalassemia trait.

Hematocrit (HCT/PCV)
M ~40–50% · F ~36–44%
 The percentage of blood volume occupied by red cells. Usually moves with hemoglobin; sensitive to hydration status.

MCH
~27–33 pg/cell
 Average hemoglobin per red cell. Read together with MCV and MCHC; usually falls alongside MCV.

MCHC
~32–36 g/dL
 Hemoglobin concentration within red cells. Low in hypochromic patterns such as iron deficiency; unexpectedly high values may reflect spherocytes or sample artefact.

DECODE THIS

Mean Corpuscular Volume MCV

WHAT IT MEASURES

The average size of red blood cells. Its real power is classification.

TYPICAL ADULT INTERVAL

~80–100 fL

▲ WHEN IT RUNS HIGHER

Above 100 fL = **macrocytic**: B12 or folate deficiency, alcohol-related changes, liver disease, hypothyroidism, certain medicines, marrow disorders.

▼ WHEN IT RUNS LOWER

Below 80 fL = **microcytic**: iron deficiency, thalassemia traits, some chronic inflammatory states.

READ IT ALONGSIDE Hb, RDW, ferritin, B12/folate, reticulocytes, peripheral smear

PEARL *MCV classifies anemia; it does not identify the cause on its own.*



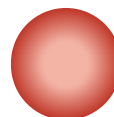
< 80 fL

Microcytic



80–100 fL

Normocytic



> 100 fL

Macrocytic

RDW — Red cell Distribution Width

~11.5–14.5%

Measures variation in red-cell size. A high RDW means the cells are mixed in size — often seen when a new population of cells (smaller or larger) is being produced, as in early nutritional deficiency.

PATTERN LAB

Pattern A — Small, varied red cells

Hb ↓

MCV ↓

RDW ↑

Ferritin ↓

THINK An iron-deficiency pattern becomes an important consideration.

ASK NEXT Why is iron depleted? Blood loss, poor intake and poor absorption are very different problems.

PATTERN LAB

Pattern B — Large red cells

Hb ↓

MCV ↑

THINK Consider causes of macrocytic anemia: B12 or folate deficiency, medication effects, alcohol- or liver-related changes, hypothyroidism and others.

ASK NEXT B12, folate, reticulocytes, liver tests, TSH, smear.

The white-cell system

DECODE THIS

White Blood Cell Count WBC / TLC

WHAT IT MEASURES

The total number of white cells — the immune system's circulating workforce.

TYPICAL ADULT INTERVAL

~4,000–11,000/μL

▲ WHEN IT RUNS HIGHER

Infection, inflammation, physiological stress, corticosteroids, tissue injury, smoking, some hematological disorders.

▼ WHEN IT RUNS LOWER

Viral illness, medication effects, marrow suppression, autoimmune conditions, sometimes severe infection.

READ IT ALONGSIDE The differential — always. A total count without the differential is half a sentence.

PEARL Absolute counts (cells/μL) are usually more informative than percentages.

Cell type	Approx. share	Often rises with	Often falls with
Neutrophils	40–70%	Bacterial infection, inflammation, stress, steroids	Medicines, viral illness, marrow suppression
Lymphocytes	20–40%	Viral infections, some chronic infections, lymphoid disorders	Acute stress, steroids, some immune deficiencies
Monocytes	2–8%	Chronic infection, inflammation, recovery phase	Marrow suppression
Eosinophils	0–6%	Allergy, asthma, parasites, drug reactions	Steroids, acute stress

Cell type	Approx. share	Often rises with	Often falls with
Basophils	0–1%	Allergic states, some myeloproliferative disorders	Rarely significant

ANC — THE NUMBER THAT MATTERS

The

ABSOLUTE NEUTROPHIL COUNT (ANC)

= $\text{WBC} \times \text{neutrophil \%}$. It is generally more clinically useful than the percentage alone, because a “normal” percentage of a very low total can still mean too few neutrophils.

The platelet system

DECODE THIS

Platelet Count PLT

WHAT IT MEASURES

Small cell fragments that plug vessel injuries and start clot formation.

TYPICAL ADULT INTERVAL

~150,000–450,000/ μL

▲ WHEN IT RUNS HIGHER

Thrombocytosis — usually reactive to infection, inflammation, iron deficiency, surgery or tissue injury; less commonly a myeloproliferative disorder.

▼ WHEN IT RUNS LOWER

Thrombocytopenia — viral infections (including dengue), medicines, immune thrombocytopenia, liver/spleen disease, marrow disorders, consumptive coagulation disorders.

READ IT ALONGSIDE Hb and WBC (is more than one line affected?), smear, coagulation profile

PEARL *Platelet clumping in the tube can produce a falsely low count. An unexpected low result is often checked on a smear or a fresh sample.*

Iron Studies — Finding the Story Behind Anemia

Iron is stored, transported, and finally used. Each test looks at a different stage.

Iron assessment should never be reduced to serum iron alone. Think of iron moving through a supply chain — each test measures a different point along it.



DECODE THIS Serum Ferritin Ferritin	
WHAT IT MEASURES An estimate of stored iron — with an important catch: ferritin is also an acute-phase reactant that rises with inflammation.	TYPICAL ADULT INTERVAL Varies considerably by sex, age and laboratory
▲ WHEN IT RUNS HIGHER Inflammation, infection, liver disease, metabolic disorders and iron overload. High ferritin does not automatically mean too much iron.	▼ WHEN IT RUNS LOWER Strongly supports depleted iron stores in the appropriate context.
READ IT ALONGSIDE Serum iron, TIBC, transferrin saturation, CRP (to judge inflammation)	
PEARL A “normal” ferritin during active inflammation may still hide iron deficiency.	

Serum iron

~60–170 µg/dL

The iron circulating at the moment of sampling. It varies through the day and after meals, so it is not a reliable standalone measure of body iron.

TIBC

~250–450 µg/dL

Total iron-binding capacity — reflects transferrin. Often rises in iron deficiency (the body makes more carrier) and falls in chronic inflammation.

Transferrin saturation

~20–50%

Serum iron ÷ TIBC × 100. Shows how loaded the carrier is. Low supports iron deficiency; persistently high values prompt evaluation for iron overload.

Reticulocyte count

~0.5–2.5%

Newly released red cells. A brisk rise shows the marrow responding to blood loss or hemolysis; a low response in anemia suggests impaired production.

Reading the iron panel as a set

Pattern	Ferritin	Serum iron	TIBC	TSAT
Iron deficiency	↓	↓	↑	↓
Anemia of inflammation	N or ↑	↓	↓ or N	↓ or N
Iron overload	↑	↑	N or ↓	↑

Typical directions only — real reports frequently show overlapping or mixed pictures.

INSTEAD OF

~~Ferritin = 8 → "low"~~

READ

Hb ↓ + MCV ↓ + Ferritin ↓ + TSAT ↓ → a much stronger iron-deficiency pattern than any single number alone

Blood Sugar — A Snapshot vs a Timeline

Glucose tells you about now. HbA1c tells you about the last few months.

GLUCOSE

The snapshot

What is happening around the time of sampling? Affected by meals, stress, illness and medication.

HbA1c

The timeline

What has glycemic exposure looked like over roughly the past 2–3 months, weighted towards recent weeks?

Unlike most tests in this book, glucose markers are interpreted against **clinical categories** rather than ordinary reference intervals. The categories below follow the American Diabetes Association criteria for non-pregnant adults.

Test	Normal	Prediabetes	Diabetes criterion
Fasting plasma glucose	<100 mg/dL	100–125 mg/dL	≥126 mg/dL
HbA1c	<5.7%	5.7–6.4%	≥6.5%
2-h glucose, 75-g OGTT	<140 mg/dL	140–199 mg/dL	≥200 mg/dL
Random plasma glucose	—	—	≥200 mg/dL with classic symptoms or hyperglycemic crisis

CONFIRMATION MATTERS

“Fasting” means no caloric intake for at least 8 hours. In the absence of unequivocal hyperglycemia, an abnormal diagnostic result needs confirmation — by repeating the same test or by a second, different test.

When HbA1c can mislead

HbA1c measures glucose attached to hemoglobin, so anything that changes red-cell lifespan or the hemoglobin molecule can shift it independently of glucose. Interpret cautiously with:

- Significant anemia, especially iron deficiency
- Recent blood loss or transfusion
- Hemolysis
- Some hemoglobin variants
- Pregnancy
- Advanced kidney disease and some of its treatments

KEY IDEA

HbA1c is not a perfect “three-month glucose average” in every person. It is a timeline drawn on red cells — and red cells can change.

PATTERN LAB

Sustained hyperglycemia

Fasting glucose ↑

HbA1c ↑

THINK The two agree: this supports persistent hyperglycemia rather than one isolated elevated reading.

PATTERN LAB

Snapshot and timeline disagree

Glucose ↑

HbA1c not raised

THINK Could the glucose rise be recent or acute — illness, stress, steroids, a non-fasting sample? Is anything lowering HbA1c?

ASK NEXT Repeat fasting glucose, consider OGTT, review red-cell factors.

Kidney Panel — Filtration, Waste & Balance

08

Never read creatinine alone. A kidney panel answers four separate questions.

FILTRATION <i>How well are the kidneys clearing the blood?</i> • Creatinine • eGFR • (Cystatin C)	WASTE PRODUCTS <i>What is accumulating?</i> • Urea • BUN	KIDNEY DAMAGE <i>Is the filter leaking?</i> • Urine albumin • ACR • Urinalysis	INTERNAL BALANCE <i>Are salts and acids controlled?</i> • Sodium • Potassium • Bicarbonate • Calcium • Phosphate
---	--	--	--

DECODE THIS Serum Creatinine Cr	
WHAT IT MEASURES A waste product of muscle metabolism, cleared mainly by kidney filtration.	TYPICAL ADULT INTERVAL ~0.6–1.3 mg/dL — strongly dependent on lab, muscle mass, age and sex
▲ WHEN IT RUNS HIGHER Reduced filtration, dehydration, high muscle mass, large meat intake, some medicines that block creatinine secretion.	▼ WHEN IT RUNS LOWER Low muscle mass, older age, frailty, pregnancy.
READ IT ALONGSIDE eGFR, urea/BUN, urine ACR, previous creatinine values	
PEARL Creatinine ≠ kidney function by itself. A frail older adult may have a “normal” creatinine with significantly reduced filtration.	

eGFR — turning creatinine into an estimate

The estimated glomerular filtration rate uses creatinine (sometimes with cystatin C), age and sex to estimate filtering capacity. KDIGO groups it into categories:

G1 ≥90 Normal or high	G2 60–89 Mildly decreased	G3a 45–59 Mild–moderate	G3b 30–44 Moderate–severe	G4 15–29 Severely decreased	G5 <15 Kidney failure
------------------------------------	--	--------------------------------------	--	--	------------------------------------

eGFR in mL/min/1.73 m²

! Don't diagnose from one result

An eGFR below 60 is important, but chronic kidney disease requires evidence that the change has lasted more than three months and/or markers of kidney damage such as albuminuria. A single value — especially during acute illness or dehydration — is not enough.

Albuminuria — the damage marker

The urine albumin-to-creatinine ratio (ACR) detects albumin leaking through the kidney filter, often before filtration falls.

A1 <30 mg/g Normal to mildly increased	A2 30–299 mg/g Moderately increased	A3 ≥300 mg/g Severely increased
---	--	--

Urea

~15–40 mg/dL

Affected by kidney function, hydration, protein intake, gastrointestinal bleeding and catabolic states — not only by the kidneys.

BUN

~7–20 mg/dL

Blood urea nitrogen; urea expressed as its nitrogen content. A disproportionately high BUN:creatinine ratio can point towards dehydration or upper GI bleeding.

PATTERN LAB

Kidney pattern

Creatinine ↑

eGFR ↓

ACR ↑

THINK Impaired filtration plus albuminuria — far more informative than creatinine alone, and deserving of clinical evaluation.

ASK NEXT Is it acute or chronic? Repeat over time and review blood pressure, glucose and medicines.

Liver Panel — Injury, Flow & Function

09

“Liver function tests” mostly don’t measure function. Sort them by the question they answer.

1

Are liver cells being injured?

ALT · AST

2

Is bile flow obstructed?

ALP · GGT · Bilirubin

3

Is the liver still making what it should?

Albumin · PT/INR

DECODE THIS

Alanine Aminotransferase ALT / SGPT

WHAT IT MEASURES

An enzyme concentrated in liver cells; released when they are injured.

TYPICAL ADULT INTERVAL

~7–56 U/L

▲ WHEN IT RUNS HIGHER

Fatty liver disease, viral hepatitis, medication or alcohol effects, autoimmune and many other liver conditions.

▼ WHEN IT RUNS LOWER

Rarely meaningful clinically.

READ IT ALONGSIDE AST, ALP, GGT, bilirubin, history of medicines and alcohol

PEARL The most liver-specific of the routine enzymes.

AST / SGOT

~10–40 U/L

Also in muscle, heart and red cells, so less liver-specific. A rise after heavy exercise or with muscle injury is common.

ALP

~44–147 U/L

From liver/bile ducts and bone (also placenta). Higher in growing children, adolescents and pregnancy.

GGT

Lab- and sex-specific

Helps decide whether a raised ALP is hepatobiliary. Also sensitive to alcohol and some medicines.

Total bilirubin

~0.2–1.2 mg/dL

A breakdown product of hemoglobin. Read the direct/indirect split rather than the total alone.

Direct bilirubin

~0–0.3 mg/dL

Conjugated fraction. A predominantly direct rise suggests liver or bile-flow problems; predominantly indirect suggests hemolysis or Gilbert syndrome.

Albumin

3.5–5.0 g/dL

Made by the liver, but also low with kidney protein loss, inflammation, malnutrition and malabsorption — not liver-specific.

Total protein

~6.0–8.3 g/dL

Albumin plus globulins. A high globulin fraction may prompt further protein studies.

PT / INR

INR ~0.8–1.2

Clotting factors are made in the liver, so a prolonged PT can reflect reduced synthetic function — or vitamin K deficiency, or anticoagulants.

Hepatocellular vs cholestatic

Hepatocellular pattern

ALT and AST rise most; ALP only mildly changed.

Think: injury to liver cells.

Cholestatic pattern

ALP and GGT rise most; ALT/AST relatively less.

Think: impaired bile formation or flow.

PATTERN LAB

Where is the ALP coming from?

ALP ↑

GGT ↑

THINK Supports a hepatobiliary source. ALP ↑ with a normal GGT shifts attention towards bone and other sources — though neither pattern diagnoses a cause on its own.

Lipid Profile — Beyond “Good” and “Bad”

A lipid report is a cardiovascular-risk puzzle, not a cholesterol scorecard.

“Good cholesterol” and “bad cholesterol” are useful shortcuts, but they oversimplify. The real question a lipid profile helps answer is: *what is this person's overall cardiovascular risk, and how much do the lipids contribute to it?*

Marker	Common descriptive level	How to read it
Total cholesterol	<200 mg/dL desirable	A blunt summary. Never assess risk with total cholesterol alone.
LDL-C	<100 mg/dL often described as optimal	The main atherogenic measure. Treatment targets depend on overall risk — high-risk people may have much lower goals.
HDL-C	Low if men <40, women <50 mg/dL	Low HDL adds to risk, but a high HDL is not a guaranteed protective score.
Triglycerides (fasting)	<150 normal · 150–199 borderline · 200–499 high · ≥500 very high	Rise after meals and with alcohol, obesity, diabetes. Very high levels raise pancreatitis risk.
Non-HDL-C	Total – HDL	Captures all atherogenic particles; useful when triglycerides are raised.
VLDL	Often calculated	Triglyceride-rich particles; usually estimated, not measured.
ApoB	Lab-specific	Counts atherogenic particles directly; may refine risk when LDL and triglycerides disagree.
ApoA1	Lab-specific	The main protein of HDL particles.
Lp(a)	Lab-specific	Largely genetic; a raised level is an independent risk-enhancing factor, usually measured once.

KEY IDEA

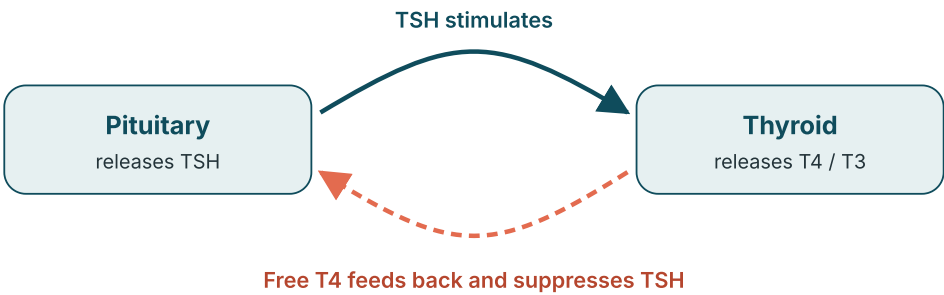
The same LDL value can be acceptable for one person and too high for another. Context — age, blood pressure, diabetes, smoking, kidney function, family history — decides what the number means.

Pieces of the risk puzzle



Thyroid Tests — Reading the Feedback Loop

TSH and free T4 make sense only as a conversation between two glands.



When the thyroid underproduces, the pituitary “shouts louder” — TSH rises. When the thyroid overproduces, TSH is suppressed. That is why the two are read together, and why TSH often moves before free T4 does.

TSH	Free T4 (± FT3)	Biochemical pattern
↑	↓	Primary hypothyroid pattern
↓	↑ (and/or FT3 ↑)	Thyrototoxic / hyperthyroid pattern
↑	Within interval	Possible subclinical hypothyroid pattern
↓	Within interval	Possible subclinical hyperthyroid pattern — check FT3 and context
↓ or inappropriately normal	↓	Consider central (pituitary/hypothalamic) causes, or non-thyroidal illness

TSH

~0.4–4.5 mIU/L

The most sensitive first-line test for primary thyroid disease. Pregnancy uses different, trimester-specific intervals.

Free T4

~0.8–1.9 ng/dL

The unbound, active fraction of thyroxine. Method-dependent; biotin can interfere with some assays.

Free T3 / Total T3

Lab-specific

Most useful when TSH is suppressed; T3 may rise before T4 in some hyperthyroid states.

Total T4

Lab-specific

Includes protein-bound hormone, so it shifts with binding-protein changes such as pregnancy or estrogen.

Anti-TPO / Anti-Tg

Negative or below lab cut-off

Antibodies that point towards autoimmune thyroid disease; positive results add context rather than a diagnosis by themselves.

Thyroglobulin

Context-specific

Mainly used to monitor people treated for differentiated thyroid cancer, not as a screening test.

TIMING CAUTION

Acute illness, recovery from illness, steroids and some other medicines can temporarily shift TSH and thyroid hormones. Abnormal results during illness are often rechecked once the person has recovered.

Electrolytes — The Body's Chemical Balance

Small numbers, tight ranges, large consequences.

Electrolytes are kept within narrow limits because nerves, muscles and the heart depend on them. Each card below follows the same order: **role** → **range** → **high** → **low** → **interferences** → **connect with**.

Na⁺

Sodium

135–145 mmol/L

Governs water distribution and cell volume.

- HIGH** Hypernatremia — water loss, inadequate intake.
- LOW** Hyponatremia — water excess, diuretics, SIADH, heart/liver/kidney conditions.
- WATCH** Very high glucose or lipids can alter measured sodium.
- CONNECT** Glucose, osmolality, fluid status, medicines

K⁺

Potassium

3.5–5.0 mmol/L

Sets the electrical stability of heart and muscle.

- HIGH** Kidney impairment, some medicines (e.g. ACE inhibitors, potassium-sparing diuretics), acidosis.
- LOW** Diuretics, vomiting, diarrhea, some hormonal states.
- WATCH** **Hemolysis**
and
delayed
processing
can falsely
raise it.
- CONNECT** Creatinine/eGFR, bicarbonate, ECG, medicines

Cl⁻

Chloride

98–106 mmol/L

Partners with sodium; helps acid-base balance.

- HIGH** Dehydration, some forms of metabolic acidosis.
- LOW** Vomiting, metabolic alkalosis.
- WATCH** Usually follows sodium.
- CONNECT** Sodium, bicarbonate (anion gap)



Bicarbonate

22–29 mmol/L

The main metabolic buffer of the blood.

- HIGH** Metabolic alkalosis, compensation for respiratory acidosis.
- LOW** Metabolic acidosis, compensation for respiratory alkalosis.
- WATCH** Under-filled tubes and delays can lower it.
- CONNECT** ABG, chloride, sodium (anion gap)



Calcium (total)

~8.5–10.5 mg/dL

Bones, nerve conduction, muscle contraction, clotting.

- HIGH** Hyperparathyroidism, malignancy, excess vitamin D.
- LOW** Low albumin, vitamin D deficiency, hypoparathyroidism, kidney disease.
- WATCH** Total calcium falls with low albumin — ionized calcium shows the active fraction.
- CONNECT** Albumin, phosphate, PTH, vitamin D



Magnesium

~1.7–2.2 mg/dL

Neuromuscular function and cardiac rhythm.

- HIGH** Kidney impairment, excess intake.
- LOW** Poor intake, alcohol, diarrhea, diuretics, proton-pump inhibitors.
- WATCH** Low magnesium can make low potassium and calcium hard to correct.
- CONNECT** Potassium, calcium



Phosphate

~2.5–4.5 mg/dL

Energy metabolism and bone mineral.

- HIGH** Kidney impairment, hypoparathyroidism.
- LOW** Malnutrition, refeeding, hyperparathyroidism, alcohol.
- WATCH** Hemolysis can raise it.
- CONNECT** Calcium, creatinine, PTH, vitamin D

A USEFUL CALCULATION

Anion gap = $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$. A raised gap in metabolic acidosis suggests an unmeasured acid (for example ketones or lactate). Typical values depend on the analyzer.

Vitamins & Nutritional Markers

Nutrition is assessed with a mosaic of tests — and many of their cut-offs are debated.

DECODE THIS

Vitamin B12Cobalamin

WHAT IT MEASURES Needed for DNA synthesis in red cells and for nerve function.	TYPICAL ADULT INTERVAL Often ~200–900 pg/mL; cut-offs vary considerably
▲ WHEN IT RUNS HIGHER Supplementation, liver disease, some myeloproliferative disorders.	▼ WHEN IT RUNS LOWER Low intake (e.g. strict vegetarian diet), pernicious anemia, malabsorption, long-term metformin or acid-suppressing medicines.
READ IT ALONGSIDE CBC/MCV, folate; methylmalonic acid or homocysteine when results are borderline	
PEARL A borderline B12 may need additional biomarkers and clinical correlation before conclusions are drawn.	

DECODE THIS

25-hydroxy vitamin D 25(OH)D

RANGE DEPENDS ON CONTEXT
WHAT IT MEASURES

The main circulating form of vitamin D, reflecting sun exposure, diet and supplements.

TYPICAL ADULT INTERVAL

Use the report's interval — definitions of sufficiency differ between organizations

▲ WHEN IT RUNS HIGHER

Excess supplementation (rarely a natural cause).

▼ WHEN IT RUNS LOWER

Limited sun exposure, darker skin, obesity, malabsorption, kidney or liver disease.

READ IT ALONGSIDE Calcium, phosphate, PTH, ALP

PEARL *There is no universal "perfect" value. Interpret by assay, clinical context and the professional guidance that applies.*

Why no numbers for some markers? Many micronutrients don't have a single trustworthy universal range. Where that is true, this book points you to the laboratory's own interval instead of inventing one.

Marker	Reference	Read it with
Folate	Method-dependent	CBC/MCV, B12, diet and alcohol history
Iron / Ferritin	See Chapter 6	Full iron panel and CRP
Calcium · Magnesium · Phosphate	See Chapter 12	Albumin, PTH, vitamin D
Zinc	Lab-specific	Albumin and inflammation (zinc falls in acute illness)
Copper	Lab-specific	Ceruloplasmin, CBC, neurological symptoms
Selenium	Lab-specific	Diet and clinical context
Vitamin A	Lab-specific	Liver function, fat malabsorption
Vitamin E	Lab-specific	Lipid levels, fat malabsorption
Vitamin K status	Assessed indirectly via PT/INR	Diet, antibiotics, malabsorption, anticoagulants
Albumin	3.5–5.0 g/dL	Inflammation, liver and kidney status

Marker	Reference	Read it with
Prealbumin	Lab-specific	Falls quickly with inflammation — not a pure nutrition marker

PART III

Specialized Investigations

Tests that are ordered for a reason — and that are especially easy to misread without context.

- Cardiac biomarkers
- Inflammation & infection
- Coagulation
- Urinalysis
- Stool & digestive tests
- Hormones
- Fertility & pregnancy
- Autoimmune tests
- Tumor markers
- Allergy & immunology
- Blood bank
- Arterial blood gas

Cardiac Biomarkers

A cardiac marker is evidence of heart stress or injury — not a diagnosis of its cause.

KEY CONCEPT

Cardiac marker ≠ cardiac diagnosis.

DECODE THIS

Troponin I / Troponin T

cTnI · cTnT · hs-cTn

RANGE DEPENDS ON CONTEXT

WHAT IT MEASURES

Proteins of the heart-muscle contractile apparatus. They enter the blood when heart muscle cells are injured.

TYPICAL ADULT INTERVAL

Assay-specific. Interpreted against that assay's 99th-percentile upper reference limit

▲ WHEN IT RUNS HIGHER

Myocardial injury from many causes: heart attack, myocarditis, heart failure, pulmonary embolism, sepsis, kidney disease, strenuous exertion.

▼ WHEN IT RUNS LOWER

Not clinically meaningful.

READ IT ALONGSIDE Symptoms, ECG, **SERIAL CHANGE** (a rise and/or fall over hours), echocardiography

PEARL An elevated troponin indicates myocardial injury but does not by itself identify the cause.

Test	What it tells you	Interpretation note
BNP / NT-proBNP	Released when heart chambers are stretched; supports evaluation of heart failure.	No universal cut-off: depends on age, kidney function, obesity (lowers levels), rhythm and acute vs chronic setting.
CK-MB	A heart-predominant fraction of creatine kinase.	Assay-specific; troponin is now more central to assessing myocardial injury.

Test	What it tells you	Interpretation note
CK (total)	Muscle enzyme from skeletal and heart muscle.	Rises with exercise, injury, intramuscular injections, some medicines (e.g. statin-related muscle effects).
Myoglobin	Small oxygen-binding muscle protein.	Rises early but is not heart-specific; limited use today.
hs-CRP	Low-grade inflammation measured with a high-sensitivity assay.	Used in cardiovascular risk assessment — not for diagnosing acute heart problems.

! Chest pain is an emergency question, not a lab question

Sudden chest pain, breathlessness or fainting needs urgent medical assessment. Troponin results are interpreted by clinicians alongside the ECG and examination, often over several hours.

Inflammation & Infection

15

Inflammation markers say “something is happening.” Infection tests try to say “what.”

CRP

Commonly <10 mg/L (labs vary)

A liver-made acute-phase protein that rises within hours of inflammation. Nonspecific: it tells you inflammation may be present, not its cause.

ESR

Varies strongly with age and sex

How quickly red cells settle. Rises with inflammation, infection, autoimmune disease, anemia, pregnancy, malignancy and older age. Slow to rise and slow to fall.

Procalcitonin

Assay- and setting-specific thresholds

Tends to rise more in bacterial than viral infection; used in specific clinical settings to support decisions, not alone.

Four ways to look for an infection

Instead of memorizing infection tests alphabetically, sort them by *what they detect*. This tells you what a positive or negative result can and cannot mean.

ANTIGEN

Looks for parts of the pathogen itself.

Dengue NS1 · HBsAg · malaria rapid antigen · some HIV combo tests

ANTIBODY

Looks for the body's immune response — which takes time to develop and may persist after recovery.

Dengue IgM/IgG · Anti-HCV · HIV antibody

MOLECULAR

Looks for the pathogen's genetic material (e.g. PCR). Highly sensitive.

HCV RNA · HIV viral load · TB NAAT

CULTURE

Tries to grow living organisms, allowing identification and antibiotic sensitivity testing.

Blood · urine · sputum cultures

Test	Type	Key interpretation point
Blood / urine / sputum culture	Culture	Collection technique matters; contamination can mimic infection. Sensitivity reports guide antibiotic choice.
Dengue NS1	Antigen	Most useful in the first days of fever; a negative result later in illness does not exclude dengue.
Dengue IgM / IgG	Antibody	IgM appears after several days; IgG patterns help distinguish primary from secondary infection.
Malaria testing	Smear / antigen / molecular	A single negative smear may not exclude malaria; repeat testing is common when suspicion persists.
HIV testing	Antigen/antibody, molecular	A reactive screening test requires confirmatory testing. Window periods apply.
HBsAg	Antigen	Indicates current hepatitis B infection; further markers determine phase.
Anti-HCV	Antibody	Shows exposure; HCV RNA determines current infection.
TB investigations	Various	Sputum tests, molecular assays, IGRA and skin tests answer different questions — latent vs active infection.

PATTERN LAB

Inflammatory picture

WBC ↑

Neutrophils ↑

CRP ↑

THINK Supports an acute inflammatory or infectious process — but does not identify an organism or prove bacterial infection.

Coagulation

16

Clotting tests measure how long it takes blood to form a clot in the tube
— and why that might change.

Test	Typical interval	What it reflects
Platelet count	~150,000–450,000/ μ L	The number of platelets available for the first plug.
PT (prothrombin time)	~11–13.5 s (method-dependent)	Extrinsic & common pathway; liver synthesis; vitamin K status.
INR	~0.8–1.2 (not on vitamin K antagonists)	PT standardized across labs — used to monitor warfarin.
aPTT	~25–35 s (reagent-dependent)	Intrinsic & common pathway; heparin effect; some factor deficiencies.
Fibrinogen	~200–400 mg/dL	The raw material of the clot; also an acute-phase protein.
Thrombin time	Lab-specific	The final conversion step; affected by heparin and fibrinogen problems.
D-dimer	Assay- and unit-specific	Fragments released when a formed clot breaks down.

INR ON TREATMENT

Therapeutic INR targets for people on anticoagulants are indication-specific (for example 2.0–3.0 for many conditions) and must never be confused with the normal reference interval.

! D-dimer: the most misread clotting test

D-dimer assays and reporting units differ, so there is no universal cut-off. A **positive** D-dimer does not diagnose a clot — it rises with infection, surgery, pregnancy, cancer, older age and inflammation. Its main strength is the opposite: a **negative** result in a low-probability setting helps make a clot less likely.

Urinalysis — A Journey Through the Report

Look, measure, test, magnify — then connect.

STEP 1

Look

Color — usually pale yellow to amber.

Appearance — usually clear. Cloudiness can come from cells, crystals, mucus or bacteria.

STEP 2

Concentration

Specific gravity ~1.005–1.030. Higher means more concentrated urine (e.g. dehydration); fixed low values may reflect impaired concentrating ability.

STEP 3

Chemistry (dipstick)

Pad	Usual	Notes
pH	4.5–8.0	Diet, infection and acid-base status influence it.
Protein	Negative–trace	Persistent protein needs quantitative ACR/UPCR.
Glucose	Negative	Appears when blood glucose exceeds the kidney threshold — or with certain medicines (SGLT2 inhibitors).
Ketones	Negative	Fasting, low-carbohydrate diets, vomiting, diabetic ketoacidosis.
Blood	Negative	Detects RBCs, hemoglobin or myoglobin — confirm with microscopy. Menstruation can contaminate.
Nitrite	Negative	Positive supports nitrate-reducing bacteria; negative does not rule out UTI.
Leukocyte esterase	Negative	Suggests white cells in urine.
Bilirubin	Negative	Conjugated bilirubin — liver/bile-flow problems.
Urobilinogen	Normal/trace	Changes with hemolysis and liver disease.

STEP 4

Microscope

RBCs often 0–2 or 0–3/HPF · **WBCs** often 0–5/HPF · **Epithelial cells** (many squamous cells suggest contamination) · **Casts** (formed in kidney tubules; some types point to kidney disease) · **Crystals** · **Bacteria** · **Yeast**.

STEP 5

Connect the pattern

Combine the findings with symptoms and the collection method.

PATTERN LAB

Urinary infection picture

Leukocyte esterase +

Nitrite +

Urine WBC ↑

THINK With compatible symptoms, supports evaluation for bacterial urinary infection; culture may be sent depending on circumstances.

SYMPTOMS DECIDE A LOT

Bacteria in urine without symptoms is common, especially in older adults, and often does not need treatment. Results are interpreted together with how the person feels.

Stool & Digestive Testing

18

Different tests for different parts of the gut: intestine, stomach and pancreas.

Most stool tests are reported as findings rather than numbers. Group them by the region they investigate.

INTESTINAL

Infection, bleeding, inflammation

- Stool routine & microscopy
- Occult blood — usual: Negative
- Ova & parasites — usual: Not detected
- Stool culture — usual: No pathogen isolated
- Fecal calprotectin — assay-specific

GASTRIC

Stomach infection

- H. pylori stool antigen
- Urea breath test
- Serology (limited role)

PANCREATIC

Digestive enzymes & inflammation

- Fecal pancreatic elastase
- Serum lipase
- Serum amylase

Fecal calprotectin

Assay-specific

A protein from neutrophils in the bowel wall. Helps separate inflammatory bowel conditions from functional symptoms; raised by infection and some medicines too.

Lipase

Relative to the lab's upper reference limit

More pancreas-specific than amylase. Interpreted against the lab's upper limit, always with symptoms and imaging where needed.

Amylase

Lab's own interval

Comes from pancreas and salivary glands; less specific than lipase. Neither enzyme should be read without symptoms and context.

Fecal occult blood

Negative

Detects hidden blood. Used in screening programmes; a positive result usually leads to further evaluation.

The Hormone Lab

For hormones, the context is the reference interval.

RANGE DEPENDS ON CONTEXT

Hormonal reference intervals may depend on:

Age

Sex

Time of day

Menstrual phase

Pregnancy

Menopausal status

Medication

Assay method

For every hormone in this chapter, use the age-, sex-, timing- and assay-specific interval printed by the performing laboratory. Printing a single universal number here would mislead.

Hormone	What it reflects	Key interpretation factor
Cortisol	Adrenal stress hormone	Strong morning peak; sample timing is critical. Steroid medicines interfere.
ACTH	Pituitary signal to the adrenals	Read with cortisol to locate the problem (adrenal vs pituitary).
Prolactin	Pituitary hormone for lactation	Raised by stress, pregnancy, some medicines; macroprolactin can cause misleading highs.
Testosterone	Main androgen	Morning sample; read with SHBG; sex- and age-specific.
Estradiol	Main estrogen	Varies across the menstrual cycle and with menopause.
Progesterone	Luteal-phase hormone	Timing in the cycle determines meaning (e.g. mid-luteal check of ovulation).
FSH / LH	Pituitary gonadotropins	Cycle-day dependent; high in menopause and primary gonadal failure.
DHEA-S	Adrenal androgen	Age-dependent; helps locate the source of excess androgens.
SHBG	Binding protein for sex hormones	Changes free vs total hormone balance; affected by liver, thyroid, weight.
AMH	Ovarian reserve marker	Reflects egg quantity, not quality; assay-dependent.

Hormone	What it reflects	Key interpretation factor
IGF-1	Growth hormone activity marker	Age-specific; more stable than growth hormone itself.
Growth hormone	Pituitary hormone	Pulsatile release — single random values are rarely useful; dynamic tests are used.
Renin / Aldosterone	Blood-pressure & salt regulation	Posture, salt intake, potassium and many blood-pressure medicines affect results.

Fertility & Pregnancy Tests

Fertility results describe a moment in a dynamic system — rarely a final verdict.

DECODE THIS

Beta human chorionic gonadotropin β -hCG

RANGE DEPENDS ON CONTEXT

WHAT IT MEASURES A hormone produced by the placenta; the basis of pregnancy testing.	TYPICAL ADULT INTERVAL Non-pregnant and pregnancy values are assay- and gestational-age dependent
▲ WHEN IT RUNS HIGHER Pregnancy (normal, multiple, or abnormal), some tumors.	▼ WHEN IT RUNS LOWER Very early pregnancy, falling levels after pregnancy loss.
READ IT ALONGSIDE Gestational dates, ultrasound, SERIAL MEASUREMENTS	
PEARL <i>In early pregnancy, how hCG changes over 48 hours can be more informative than one isolated value.</i>	

Hormone tests used in fertility evaluation — FSH, LH, estradiol, progesterone, AMH, prolactin and testosterone — are covered in Chapter 19. Their timing in the menstrual cycle is essential to interpretation.

The semen analysis spread

Volume

Amount of ejaculate; affected by abstinence time and incomplete collection.

Sperm concentration

Sperm per milliliter.

Total sperm number

Concentration × volume — the total in the sample.

Total motility

Percentage of sperm moving in any way.

Progressive motility

Percentage moving forward actively.

Morphology

Percentage with typical shape under strict criteria.

Vitality

Percentage of live sperm — useful when motility is low.

pH

Reflects contributions from the accessory glands.

READ THE REFERENCE LIMITS WISELY

Laboratories report against current WHO-based reference limits. These limits describe men whose partners conceived — they are not hard fertility thresholds. A single parameter outside a reference limit is not synonymous with infertility, and results commonly vary between samples, so repeat testing is standard.

Autoimmune Investigation

An antibody is evidence. It is not automatically a diagnosis.

GOLDEN RULE

Positive autoimmune antibody ≠ automatic autoimmune disease.

Results are reported as **negative / positive**, as a **titer** (for example 1:160) or as a **quantitative value**, depending on the assay. They gain meaning only alongside symptoms, examination and other laboratory evidence.

Test	Associated with	Interpretation note
ANA	Screening for systemic autoimmune disease (e.g. lupus)	Low titers are common in healthy people; the pattern and titer matter.
Anti-dsDNA	Lupus	More specific than ANA; may track disease activity in some patients.
Anti-Sm	Lupus	Highly specific but not sensitive.
SSA/Ro · SSB/La	Sjögren's syndrome, lupus	Important in pregnancy planning (neonatal effects).
RF	Rheumatoid arthritis	Also positive in infections, other autoimmune conditions and older age.
Anti-CCP	Rheumatoid arthritis	More specific than RF; may appear before symptoms.
ANCA	Small-vessel vasculitis	Pattern (c/p) and specific targets (PR3/MPO) guide interpretation.
C3 / C4	Complement consumption	Low levels can accompany active immune-complex disease.
Antiphospholipid antibodies	Antiphospholipid syndrome	Need persistence on repeat testing at least 12 weeks apart plus clinical events.
HLA-B27	Spondyloarthritis	A genetic marker; common in the general population — supports, never confirms.

Tumor Markers

Useful tools in the right setting – and powerful sources of anxiety in the wrong one.

TUMOR MARKER ≠ CANCER TEST

A **high** tumor marker does not automatically mean cancer.

A **normal** tumor marker does not automatically exclude cancer.

Their usefulness depends on the marker, the cancer type, the clinical setting and the purpose of testing. Most are best at monitoring an already-diagnosed condition.

Marker	Main clinical use	Can also rise with (examples)
PSA	Prostate cancer screening discussions and monitoring	Prostate enlargement, prostatitis, recent ejaculation or cycling, procedures
Free PSA	Refining interpretation of a borderline PSA	Same factors as PSA
AFP	Liver cancer surveillance; certain germ-cell tumors	Hepatitis, cirrhosis, pregnancy
CEA	Monitoring colorectal and some other cancers	Smoking, inflammatory bowel disease, liver disease
CA-125	Monitoring ovarian cancer	Menstruation, endometriosis, pregnancy, fluid in abdomen
CA 19-9	Monitoring pancreatic/biliary cancers	Bile-duct obstruction, pancreatitis
CA 15-3	Monitoring breast cancer	Liver disease, some benign breast conditions
β-hCG	Germ-cell and trophoblastic tumors	Pregnancy
Calcitonin	Medullary thyroid cancer	Kidney disease, some medicines
Thyroglobulin	Monitoring after thyroid cancer treatment	Any thyroid tissue; antibodies can interfere

Same lab, same assay: interpret tumor markers as trends using the same assay — a change of assay can change the number without any change in the patient.

Allergy & Immunology

Sensitization is not the same thing as allergy.

KEY CONCEPT

Sensitization ≠ Clinical Allergy. A positive specific IgE shows the immune system recognizes an allergen — whether it causes symptoms is a clinical question.

Test	What it shows	Interpretation note
Total IgE	Overall IgE level	Raised in allergic disease, parasitic infection and some other conditions; not allergen-specific.
Specific IgE	IgE against a named allergen	Interpret by assay class/concentration and the person's history of reactions.
IgG · IgA · IgM	Main antibody classes	Low levels may suggest immune deficiency; high levels occur with chronic infection, inflammation or plasma-cell disorders. IgA deficiency affects some celiac tests.
Complement C3 · C4	Complement system proteins	Low with consumption in immune-complex disease; C4 also used in hereditary angioedema work-up.
Immunofixation	Identifies a monoclonal protein	Used when a plasma-cell disorder is suspected; reported as present/absent with type.

All of these use laboratory-specific intervals. Food-specific IgG tests marketed for "intolerance" are not recommended for diagnosing food allergy.

Blood Bank Reports

24

Compatibility is confirmed in the laboratory — not on a chart.

ABO group**A · B · AB · O**

Antigens on red cells, with matching naturally occurring antibodies in plasma against the antigens a person lacks.

Rh(D)**Positive · Negative**

Presence of the D antigen. Important in transfusion and pregnancy.

Antibody screen**Negative · Positive**

Searches the plasma for other red-cell antibodies, often formed after transfusion or pregnancy. Positive results need identification.

Crossmatch**Compatible · Incompatible**

Tests the patient's plasma against the specific unit to be transfused.

Direct antiglobulin test (DAT)**Negative · Positive**

Detects antibody or complement already coating the person's red cells — used in hemolysis work-up and transfusion reactions.

! Beyond the ABO chart

Simplified donor/recipient charts are useful for learning, but actual transfusion decisions require formal blood-bank testing: grouping, antibody screening and crossmatching.

Arterial Blood Gas — An Algorithm

Read an ABG in order, one step at a time.

Parameter	Typical arterial value
pH	7.35–7.45
PaCO ₂	35–45 mmHg
HCO ₃ [−]	22–26 mmol/L
PaO ₂	~80–100 mmHg in healthy younger adults at sea level — strongly influenced by age, altitude and oxygen therapy

1

pH

<7.35 →
acidemia · >7.45
→ **alkalemia**

2

PaCO₂

The respiratory component.
Does it explain the pH change?

3

HCO₃[−]

The metabolic component.
Does it explain the pH change?

4

Compensation

Is the other system responding appropriately — or is there a second disorder?

5

Oxygenation

Interpret PaO₂ and saturation in the context of inspired oxygen.

Primary disorder	pH	Primary change	Expected compensation	Example causes
Metabolic acidosis	↓	HCO ₃ [−] ↓	PaCO ₂ ↓	Diabetic ketoacidosis, lactic acidosis, kidney failure, diarrhea
Metabolic alkalosis	↑	HCO ₃ [−] ↑	PaCO ₂ ↑	Vomiting, diuretics
Respiratory acidosis	↓	PaCO ₂ ↑	HCO ₃ [−] ↑	Hypoventilation, severe COPD, sedation
Respiratory alkalosis	↑	PaCO ₂ ↓	HCO ₃ [−] ↓	Anxiety, pain, early sepsis, high altitude

Real life is messier: mixed disorders are common. The anion gap, electrolytes, history and clinical status complete the picture.

PART IV

Stop Reading Tests One by One

Ten core combinations that turn scattered flags into a clear direction.

- Microcytic & macrocytic anemia
- Kidney pattern
- Hepatocellular vs cholestatic
- Thyroid pattern
- Inflammation
- Diabetes
- Urinary infection
- Dehydration

The Pattern Lab

Each case shows a combination, the direction it points, and the question that should come next.

A pattern is not a diagnosis. It is a **direction**: a way of narrowing what is likely and choosing the next useful question. Notice that every case below ends with a question, not a conclusion.

CASE 01

Microcytic anemia pattern

Hb ↓

MCV ↓

Ferritin ↓

TSAT ↓

PATTERN DIRECTION Iron deficiency becomes an important consideration.

NEXT QUESTION Why is iron depleted? Identifying iron deficiency is not the same as identifying its cause — blood loss, low intake, poor absorption and increased need are very different stories.

CASE 02

Macrocytic pattern

Hb ↓

MCV ↑

PATTERN DIRECTION Points towards causes of large red cells.

NEXT QUESTION Look next at B12, folate, reticulocytes, liver markers, TSH, medication and alcohol history, and the peripheral smear.

CASE 03

Kidney pattern

Creatinine ↑

eGFR ↓

ACR ↑

PATTERN DIRECTION Impaired filtration plus albuminuria deserves clinical evaluation.

NEXT QUESTION Is this acute or chronic? Chronicity and context determine classification.

CASE 04

Hepatocellular pattern

ALT ↑↑

AST ↑

ALP mildly changed

PATTERN DIRECTION A liver-cell injury pattern.

NEXT QUESTION What is injuring the liver — fat, viruses, medicines, alcohol, autoimmunity, or something else?

CASE 05

Cholestatic pattern

ALP ↑↑

GGT ↑

ALT/AST less raised

PATTERN DIRECTION A cholestatic / hepatobiliary pattern.

NEXT QUESTION Is bile flow blocked outside the liver or impaired within it? Bilirubin fractions and imaging help.

CASE 06

Thyroid pattern

TSH ↑

FT4 ↓

PATTERN DIRECTION Primary hypothyroid biochemical pattern.

NEXT QUESTION What is the cause (e.g. autoimmune thyroiditis — check anti-TPO), and what is the clinical significance?

CASE 07

Infection / inflammation pattern

WBC ↑

Neutrophils ↑

CRP ↑

PATTERN DIRECTION Supports an acute inflammatory or infectious process.

NEXT QUESTION Where is the source? These markers don't identify an organism or prove bacterial infection.

CASE 08

Diabetes pattern

Fasting glucose ↑

HbA1c ↑

PATTERN DIRECTION Sustained hyperglycemia rather than an isolated reading.

NEXT QUESTION Have diagnostic criteria been met and confirmed? Check kidney function, urine ACR and lipids.

CASE 09

UTI pattern

Leukocyte esterase +

Nitrite +

Urine WBC ↑

PATTERN DIRECTION With compatible symptoms, supports evaluation for bacterial urinary infection.

NEXT QUESTION Are there symptoms? Is a culture needed for this person?

CASE 10

Possible dehydration pattern

Hb/HCT relatively ↑

Urea/BUN ↑

Urine SG ↑

PATTERN DIRECTION Could fit concentration from fluid loss in the right context.

NEXT QUESTION Each marker has other explanations — does the picture normalize after rehydration?

THE RULE OF THE LAB

Pattern → direction → next question. Never pattern → diagnosis.

PART V

Read a Report With Me

Ten fictional reports, read slowly — the way an experienced interpreter would.

- CBC
- Iron profile
- Diabetes panel
- Kidney panel
- Liver panel
- Lipid profile
- Thyroid profile
- Urinalysis
- Vitamin & nutrition panel
- Comprehensive health panel

The Report Rooms

Cover the interpretation, try the three looks yourself, then compare.

Every Report Room follows the same rhythm. Try to answer each “look” before reading the explanation.

First look What is flagged?	Second look Which system?	Third look Do related results agree?	Interpretation	Don't jump to...	What next?
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All patients, values and laboratories are fictional. Reference intervals shown are illustrative.

Complete Blood Count

SAMPLE LABORATORY

Fictional report for learning only

Female • 34 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Hemoglobin	9.8	g/dL	12.0–15.0	L
RBC	4.3	million/ μ L	4.1–5.1	
Hematocrit	31	%	36–44	L
MCV	72	fL	80–100	L
MCH	22.8	pg	27–33	L
MCHC	31.0	g/dL	32–36	L
RDW	17.2	%	11.5–14.5	H
WBC	6,800	/ μ L	4,000–11,000	
Platelets	438,000	/ μ L	150,000–450,000	

FIRST LOOK**Which values are flagged?**

Hb, HCT, MCV, MCH and MCHC are low; RDW is high. WBC and platelets are within range.

SECOND LOOK**Which system is affected?**

Only the red-cell system. White cells are normal; platelets are at the upper end of normal.

THIRD LOOK**Do related parameters support one pattern?**

Yes: small (microcytic), pale (hypochromic) cells of varying size (high RDW). The upper-normal platelets can accompany iron deficiency.

INTERPRETATION

A microcytic, hypochromic anemia with increased size variation — a pattern in which iron deficiency is an important consideration.

DON'T JUMP TO

“Iron deficiency” as a confirmed diagnosis — or a cause. Thalassemia trait and chronic inflammation can also give small cells.

WHAT COULD COME NEXT?

- Ferritin, serum iron, TIBC, transferrin saturation
- CRP if inflammation is possible
- History of menstrual or other blood loss and diet
- Peripheral smear; hemoglobin studies if iron studies are normal

SAMPLE LABORATORY

Fictional report for learning only

Male • 58 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Hemoglobin	11.6	g/dL	13.0–17.0	L
MCV	86	fL	80–100	
Ferritin	410	ng/mL	30–400	H
Serum iron	42	µg/dL	60–170	L
TIBC	220	µg/dL	250–450	L
Transferrin saturation	19	%	20–50	L
CRP	38	mg/L	<10	H

FIRST LOOK

Which values are flagged?

Hb low; ferritin and CRP high; serum iron, TIBC and saturation low.

SECOND LOOK

Which system is affected?

Red cells — with a strong inflammatory signal alongside.

THIRD LOOK

Do related parameters support one pattern?

Low iron with *low* TIBC and high ferritin is the classic shape of iron being locked away during inflammation, rather than true depletion.

INTERPRETATION

A normocytic anemia with an iron profile typical of anemia of inflammation (chronic disease). The high ferritin here is most likely driven by inflammation, not iron overload.

DON'T JUMP TO

"Iron overload" because ferritin is high, or "iron deficiency" because serum iron is low. Neither number alone tells the story.

WHAT COULD COME NEXT?

- Identify the source of inflammation
- Kidney function
- Reticulocyte count
- Consider coexisting iron deficiency if clinical suspicion remains

SAMPLE LABORATORY

Fictional report for learning only

Male • 46 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Fasting plasma glucose	134	mg/dL	70–99	H
HbA1c	6.8	%	<5.7	H
Hemoglobin	14.6	g/dL	13.0–17.0	
Creatinine	0.9	mg/dL	0.7–1.3	
eGFR	98	mL/min/1.73m ²	≥90	
Urine ACR	42	mg/g	<30	H

FIRST LOOK

Which values are flagged?

Fasting glucose, HbA1c and urine ACR.

SECOND LOOK

Which system is affected?

Glucose metabolism, with an early kidney signal.

THIRD LOOK

Do related parameters support one pattern?

Yes. Two different glucose tests both exceed diabetes criteria, and hemoglobin is normal, so HbA1c is likely reliable here.

INTERPRETATION

Fasting glucose ≥126 mg/dL and HbA1c ≥6.5% in the same sample meet diabetes criteria in a non-pregnant adult. Moderately increased albuminuria (A2) is present with normal filtration.

DON'T JUMP TO

Conclusions about kidney disease from one ACR — albuminuria needs repeat testing, as exercise, infection and high glucose can raise it temporarily.

WHAT COULD COME NEXT?

- Clinical review and diabetes care plan
- Repeat urine ACR (2 of 3 samples over 3–6 months)
- Lipid profile, blood pressure
- Eye and foot assessment as part of routine care

Kidney Panel

SAMPLE LABORATORY

Fictional report for learning only

Female • 71 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Creatinine	1.4	mg/dL	0.6–1.1	H
eGFR	38	mL/min/1.73m ²	≥90	L
Urea	58	mg/dL	15–40	H
Sodium	139	mmol/L	135–145	
Potassium	5.3	mmol/L	3.5–5.0	H
Bicarbonate	21	mmol/L	22–29	L
Urine ACR	185	mg/g	<30	H

FIRST LOOK**Which values are flagged?**

Creatinine, urea, potassium and ACR high; eGFR and bicarbonate low.

SECOND LOOK**Which system is affected?**

Kidney filtration and kidney damage markers, plus balance (potassium, acid-base).

THIRD LOOK**Do related parameters support one pattern?**

Yes: reduced filtration (G3b), albuminuria (A2), mild potassium retention and a slightly low bicarbonate all fit reduced kidney function.

INTERPRETATION

Reduced estimated filtration with albuminuria and early balance changes. Whether this is chronic kidney disease depends on whether it persists beyond three months.

DON'T JUMP TO

A CKD stage from a single test — this could be an acute change (dehydration, medicines, illness). Also don't ignore the potassium.

WHAT COULD COME NEXT?

- Compare with previous creatinine values
- Repeat tests to establish chronicity
- Medication review (NSAIDs, ACE inhibitors/ARBs, diuretics)
- Check whether potassium was affected by hemolysis
- Blood pressure, glucose, urinalysis

SAMPLE LABORATORY

Fictional report for learning only

Male • 39 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
ALT	182	U/L	7–56	H
AST	121	U/L	10–40	H
ALP	118	U/L	44–147	
GGT	74	U/L	8–61	H
Total bilirubin	1.0	mg/dL	0.2–1.2	
Albumin	4.3	g/dL	3.5–5.0	
INR	1.0		0.8–1.2	

FIRST LOOK

Which values are flagged?

ALT, AST and GGT.

SECOND LOOK

Which system is affected?

Liver — the injury markers.
Flow markers (ALP, bilirubin)
and function markers
(albumin, INR) are normal.

THIRD LOOK

Do related parameters
support one pattern?

ALT dominates over ALP,
giving a hepatocellular pattern;
synthetic function is
preserved.

INTERPRETATION

A hepatocellular injury pattern with preserved liver function. The cause is not identified by these numbers.

DON'T JUMP TO

"Fatty liver" or "alcohol" as a label without
history, examination and further tests.

WHAT COULD COME NEXT?

- History: alcohol, medicines, supplements, weight, recent illness
- Hepatitis B and C serology
- Repeat panel to see the trend
- Ultrasound if clinically indicated
- Glucose and lipids (metabolic risk)

Lipid Profile

SAMPLE LABORATORY

Fictional report for learning only

Female • 52 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Total cholesterol	236	mg/dL	<200	H
LDL-C	148	mg/dL	<100	H
HDL-C	44	mg/dL	>50	L
Triglycerides	220	mg/dL	<150	H
Non-HDL-C	192	mg/dL	<130	H

FIRST LOOK**Which values are flagged?**

Every line.

SECOND LOOK**Which system is affected?**

Lipid metabolism — which feeds into cardiovascular risk.

THIRD LOOK**Do related parameters support one pattern?**

High triglycerides with low HDL and raised non-HDL-C is a common mixed pattern, often linked with insulin resistance.

INTERPRETATION

A mixed dyslipidemia pattern. How much this matters depends on the person's whole cardiovascular risk profile.

DON'T JUMP TO

A fixed "treat or don't treat" judgment from lipid numbers alone. Also check whether the sample was truly fasting — triglycerides rise after meals.

WHAT COULD COME NEXT?

- Cardiovascular risk assessment (BP, smoking, family history, diabetes)
- HbA1c / fasting glucose
- TSH, kidney and liver tests (secondary causes)
- Consider ApoB or Lp(a) where available

Thyroid Profile

SAMPLE LABORATORY

Fictional report for learning only

Female • 44 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
TSH	9.6	mIU/L	0.4–4.5	H
Free T4	0.7	ng/dL	0.8–1.9	L
Anti-TPO	286	IU/mL	<35	H

FIRST LOOK**Which values are flagged?**

TSH high, free T4 low, anti-TPO high.

SECOND LOOK**Which system is affected?**

The pituitary–thyroid feedback loop.

THIRD LOOK**Do related parameters support one pattern?**

Yes: the pituitary is signaling harder (high TSH) because thyroid output is low, and antibodies suggest an autoimmune process.

INTERPRETATION

A primary hypothyroid biochemical pattern with positive thyroid peroxidase antibodies — consistent with an autoimmune thyroid process.

DON'T JUMP TO

Interpreting the numbers without checking recent illness, biotin supplements, or pregnancy.

WHAT COULD COME NEXT?

- Clinical review of symptoms
- Confirm results if clinically unexpected
- Lipids and CBC as appropriate
- Pregnancy plans — thyroid targets differ in pregnancy

SAMPLE LABORATORY

Female • 27 y • midstream sample

Fictional report for learning only

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Appearance	Cloudy		Clear	*
Specific gravity	1.020		1.005–1.030	
pH	6.5		4.5–8.0	
Protein	Trace		Negative	*
Blood	1+		Negative	*
Nitrite	Positive		Negative	*
Leukocyte esterase	2+		Negative	*
WBC	25–30	/HPF	0–5	H
RBC	4–6	/HPF	0–3	H
Squamous epithelial cells	Few		Few	
Bacteria	Many		None/few	*

FIRST LOOK

Which values are flagged?

Cloudy appearance, trace protein, blood, nitrite, leukocyte esterase, raised WBC and RBC, many bacteria.

SECOND LOOK

Which system is affected?

The urinary tract.

THIRD LOOK

Do related parameters support one pattern?

Yes — nitrite, leukocyte esterase, pyuria and bacteria point the same way, and few epithelial cells suggest a reasonably clean sample.

INTERPRETATION

With compatible symptoms (burning, frequency), this supports evaluation for bacterial urinary tract infection. Trace protein and some red cells often accompany infection.

DON'T JUMP TO

A diagnosis without symptoms — or ignoring persistent blood once any infection has settled.

WHAT COULD COME NEXT?

- Correlate with symptoms
- Urine culture and sensitivity where indicated
- Repeat urinalysis after recovery if blood persists

SAMPLE LABORATORY

Fictional report for learning only

Male • 63 y

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Hemoglobin	11.9	g/dL	13.0–17.0	L
MCV	108	fL	80–100	H
Vitamin B12	168	pg/mL	200–900	L
Folate	9.2	ng/mL	>4.0	
25(OH) vitamin D	17	ng/mL	lab interval 30–100	L
Calcium	9.1	mg/dL	8.5–10.5	
Albumin	4.0	g/dL	3.5–5.0	

FIRST LOOK

Which values are flagged?

Hb low, MCV high, B12 low, vitamin D below the lab's interval.

SECOND LOOK

Which system is affected?

Red cells and nutritional status.

THIRD LOOK

Do related parameters support one pattern?

Macrocytic anemia with low B12 and normal folate supports a B12-related cause.

INTERPRETATION

A macrocytic anemia with low B12, plus low vitamin D according to this laboratory's interval.

DON'T JUMP TO

A cause — B12 deficiency has many origins (diet, absorption problems, autoimmune gastritis, long-term metformin or acid-suppressing medicines).

WHAT COULD COME NEXT?

- Medication and diet history
- Consider intrinsic-factor antibodies and absorption assessment
- Neurological symptoms review
- Clinical decision about vitamin D according to applicable guidance

Comprehensive Health Panel

SAMPLE LABORATORY

Fictional report for learning only

Male • 50 y • annual check

TEST	RESULT	UNIT	REFERENCE INTERVAL	FLAG
Hemoglobin	15.1	g/dL	13.0–17.0	
WBC	7,400	/μL	4,000–11,000	
Fasting glucose	112	mg/dL	70–99	H
HbA1c	6.0	%	<5.7	H
Creatinine	1.0	mg/dL	0.7–1.3	
ALT	64	U/L	7–56	H
AST	38	U/L	10–40	
Triglycerides	245	mg/dL	<150	H
HDL-C	36	mg/dL	>40	L
LDL-C	128	mg/dL	<100	H
TSH	2.1	mIU/L	0.4–4.5	

FIRST LOOK

Which values are flagged?

Glucose and HbA1c (prediabetes range), mildly raised ALT, high triglycerides and LDL, low HDL.

SECOND LOOK

Which systems are affected?

Glucose, lipids and liver — three systems at once.

THIRD LOOK

Do related parameters support one pattern?

Together they form a recognizable metabolic picture: insulin resistance often links borderline glucose, high triglycerides, low HDL and mild ALT rises.

INTERPRETATION

Prediabetes-range glucose markers with a mixed dyslipidemia and a mild hepatocellular enzyme rise. Individually mild, together they form a metabolic risk pattern worth discussing.

DON'T JUMP TO

Dismissing each value as “only slightly high.” This is exactly where pattern reading earns its place.

WHAT COULD COME NEXT?

- Clinical review: weight, waist, blood pressure, lifestyle
- Repeat glucose markers as advised
- Liver assessment if ALT persists
- Overall cardiovascular risk estimate

PART VI

Your Lab Report Toolkit

Preparation, sample quality, abbreviations, quick-reference tables and a one-page roadmap.

- Before the test
- When the sample changes the result
- Laboratory language decoded
- Quick reference
- One-page roadmap

Before the Test

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Good results start before the needle — with preparation.

Fasting

For fasting glucose and many lipid panels: no calories for 8–12 hours as instructed. Plain water is usually allowed.

Water intake

Normal hydration helps accurate results and easier sampling — unless told otherwise.

Morning sampling

Cortisol, testosterone and some other tests depend on early-morning collection.

Medication disclosure

Tell the lab and doctor what you take. Don't stop medicines without advice — some tests are timed around doses.

Supplement disclosure

Biotin and other supplements can interfere with some assays. Ask whether to pause them before testing.

Exercise

Avoid strenuous exercise for 24–48 hours before tests such as CK, liver enzymes and urine protein.

Alcohol

Can affect triglycerides, GGT, liver enzymes and glucose. Avoid before testing if advised.

Smoking

Can raise hemoglobin, WBC and CEA, among others.

Menstrual cycle

Many hormone tests must be timed to a cycle day. Menstruation can contaminate urine samples.

Timed hormone tests

Follow the exact timing — the reference interval assumes it.

24-hour urine

Discard the first morning void, then collect every drop for 24 hours, ending with the next morning's first void. Keep cool if instructed.

Midstream urine

Clean the area, pass the first portion into the toilet, collect the middle portion, finish in the toilet.

When the Sample Changes the Result

A result can be wrong before analysis even begins. This is the pre-analytical phase.

If a result is unexpected, doesn't fit the clinical picture and could plausibly be a sample problem, the most sensible next step is often a careful repeat collection.

Problem	What happens	Commonly affected
Hemolysis	Red cells rupture in the tube and release their contents	Potassium ↑, LDH ↑, AST ↑, phosphate ↑; some assays disturbed
Lipemia	Fatty, milky sample after a meal or with very high triglycerides	Optical assays; sodium (some methods)
Icterus	Very high bilirubin colors the sample	Some colorimetric assays
Clotted specimen	Clot forms in an anticoagulated tube	Platelets ↓, CBC unreliable, coagulation tests invalid
Insufficient quantity	Wrong blood-to-additive ratio	Coagulation tests, bicarbonate
Contamination	IV fluid, skin flora or wrong-site sampling	Glucose, electrolytes, cultures
Delayed processing	Cells keep metabolizing in the tube	Glucose ↓, potassium ↑, phosphate ↑
Incorrect tube	Wrong additive (e.g. EDTA carry-over)	Calcium ↓, potassium ↑ (EDTA), others
Improper storage	Wrong temperature or light exposure	Bilirubin, some vitamins, hormones
Patient identification error	Sample belongs to someone else	Everything
Prolonged tourniquet	Local blood concentration and fist clenching	Potassium, protein, calcium

Laboratory Language Decoded 30

The abbreviations you are most likely to meet on a report.

ABG	Arterial blood gas
ACR	Albumin-to-creatinine ratio
ACTH	Adrenocorticotrophic hormone
AFP	Alpha-fetoprotein
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase (SGPT)
AMH	Anti-Müllerian hormone
ANA	Antinuclear antibody
ANC	Absolute neutrophil count
ANCA	Anti-neutrophil cytoplasmic antibody
Anti-CCP	Anti-cyclic citrullinated peptide
Anti-dsDNA	Anti-double-stranded DNA
Anti-Tg	Anti-thyroglobulin antibody
Anti-TPO	Anti-thyroid peroxidase antibody
ApoA1 / ApoB	Apolipoprotein A1 / B
aPTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase (SGOT)
β-hCG	Beta human chorionic gonadotropin
BNP	B-type natriuretic peptide
BUN	Blood urea nitrogen
C3 / C4	Complement components 3 and 4
Ca	Calcium

CA-125	Cancer antigen 125
CBC	Complete blood count
CEA	Carcinoembryonic antigen
CK	Creatine kinase
CK-MB	Creatine kinase, MB fraction
CKD	Chronic kidney disease
Cl	Chloride
Cr	Creatinine
CRP	C-reactive protein
DAT	Direct antiglobulin test
DHEA-S	Dehydroepiandrosterone sulfate
DLC	Differential leukocyte count
eGFR	Estimated glomerular filtration rate
ESR	Erythrocyte sedimentation rate
FBS / FPG	Fasting blood sugar / fasting plasma glucose
FOBT	Fecal occult blood test
FSH	Follicle-stimulating hormone
FT3 / FT4	Free triiodothyronine / free thyroxine
GGT	Gamma-glutamyl transferase
GH	Growth hormone
Hb	Hemoglobin
HbA1c	Glycated hemoglobin
HBsAg	Hepatitis B surface antigen
HCO₃⁻	Bicarbonate

HCT / PCV	Hematocrit / packed cell volume
HCV	Hepatitis C virus
HDL-C	High-density lipoprotein cholesterol
HLA-B27	Human leukocyte antigen B27
HPF	High-power field (microscopy)
hs-CRP	High-sensitivity C-reactive protein
hs-cTn	High-sensitivity cardiac troponin
IgA/IgE/IgG/IgM	Immunoglobulins A, E, G, M
IGF-1	Insulin-like growth factor 1
IGRA	Interferon-gamma release assay
INR	International normalized ratio
K	Potassium
LDH	Lactate dehydrogenase
LDL-C	Low-density lipoprotein cholesterol
LFT	Liver function tests
LH	Luteinizing hormone
Lp(a)	Lipoprotein(a)
MCH	Mean corpuscular hemoglobin
MCHC	Mean corpuscular hemoglobin concentration
MCV	Mean corpuscular volume
Mg	Magnesium
MPV	Mean platelet volume
Na	Sodium
NS1	Non-structural protein 1 (dengue antigen)

NT-proBNP	N-terminal pro-B-type natriuretic peptide
OGTT	Oral glucose tolerance test
PaCO₂	Arterial partial pressure of CO ₂
PaO₂	Arterial partial pressure of O ₂
PCR	Polymerase chain reaction
PLT	Platelets
PO₄	Phosphate
PPBS	Post-prandial blood sugar
PSA	Prostate-specific antigen
PT	Prothrombin time
PTH	Parathyroid hormone
RA factor / RF	Rheumatoid factor
RBC	Red blood cell (count)
RBS	Random blood sugar
RDW	Red cell distribution width
Retic	Reticulocyte count
SG	Specific gravity
SHBG	Sex hormone-binding globulin
SGOT / SGPT	Older names for AST / ALT
T3 / T4	Triiodothyronine / thyroxine
TG	Triglycerides
TIBC	Total iron-binding capacity
TLC	Total leukocyte count
TSAT	Transferrin saturation
TSH	Thyroid-stimulating hormone
TT	Thrombin time
UPCR	Urine protein-to-creatinine ratio
VLDL	Very-low-density lipoprotein
WBC	White blood cell (count)
25(OH)D	25-hydroxy vitamin D

Quick Reference & One-Page Roadmap

Tear-out pages for the desk, the clinic or the phone.

Which tests go together?

Anemia

CBC · MCV/RDW ·
Reticulocytes · Ferritin ·
Iron/TIBC · B12 · Folate ·
Peripheral smear

Diabetes

Fasting glucose · HbA1c ·
OGTT when indicated · Urine
ACR · Creatinine/eGFR · Lipid
profile

Kidney

Creatinine · eGFR · Urea/BUN
· Electrolytes · Urinalysis ·
ACR/UPCR

Liver

ALT · AST · ALP · GGT ·
Bilirubin · Albumin · PT/INR

Thyroid

TSH · FT4 · ± FT3 · ± thyroid
antibodies when indicated

Cardiovascular risk

Lipid profile · Glucose/HbA1c
· Kidney assessment · other
markers by indication

Quick reference table

Test	Typical adult reference / interpretation	Test	Typical adult reference / interpretation
Hb – men	~13–17 g/dL	Calcium	~8.5–10.5 mg/dL
Hb – women	~12–15 g/dL	Magnesium	~1.7–2.2 mg/dL
WBC	~4,000–11,000/μL	Phosphate	~2.5–4.5 mg/dL
Platelets	~150,000–450,000/μL	Albumin	~3.5–5.0 g/dL
MCV	~80–100 fL	Total protein	~6.0–8.3 g/dL
MCH	~27–33 pg	Total bilirubin	~0.2–1.2 mg/dL
MCHC	~32–36 g/dL	Fasting glucose	<100 mg/dL*
Sodium	~135–145 mmol/L	HbA1c	<5.7%*
Potassium	~3.5–5.0 mmol/L	2-h OGTT	<140 mg/dL*
Chloride	~98–106 mmol/L	TSH	~0.4–4.5 mIU/L

Test	Typical adult reference / interpretation	Test	Typical adult reference / interpretation
Bicarbonate	~22–29 mmol/L	Free T4	~0.8–1.9 ng/dL
ABG pH	7.35–7.45	Urine SG	~1.005–1.030
PaCO₂	35–45 mmHg	Urine pH	~4.5–8.0
HCO₃⁻ (ABG)	22–26 mmol/L	INR (no VKA)	~0.8–1.2

*Clinical categories rather than ordinary reference intervals (ADA): prediabetes = FPG 100–125 mg/dL, HbA1c 5.7–6.4% or 2-h OGTT 140–199 mg/dL; diabetes criteria = FPG ≥126 mg/dL, HbA1c ≥6.5% or 2-h OGTT ≥200 mg/dL, confirmed unless hyperglycemia is unequivocal. All other values are approximate educational intervals — the reporting laboratory's interval takes priority.

WHEN YOU RECEIVE A LAB REPORT

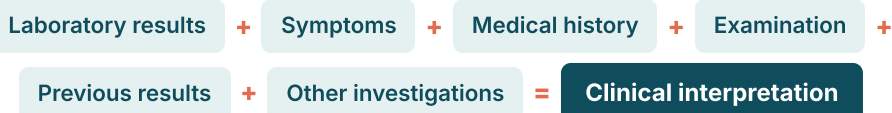
- 01 Check the test, result, unit and laboratory reference interval.
- 02 Identify the abnormal or flagged results.
- 03 Group related markers by system.
- 04 Look for a pattern.
- 05 Compare with previous results.
- 06 Consider fasting, hydration, medication, timing and sample factors.
- 07 Connect results with symptoms and clinical history.
- 08 Decide whether confirmation or further investigation may be needed.
- 09 Discuss clinically significant abnormalities with the appropriate healthcare professional.

The final checklist

- | | |
|--|--|
| ☐ Is the result truly outside <i>this</i> lab's interval? | ☐ Are the units correct? |
| ☐ Was the person properly prepared? | ☐ Could medicines or supplements affect it? |
| ☐ Could collection or handling have affected it? | ☐ What do the related tests show? |
| ☐ Is it new, or a long-term trend? | ☐ Does it fit the symptoms and history? |
| ☐ Does it need repeat or confirmatory testing? | ☐ Does it need routine or urgent review? |

THE 5-QUESTION METHOD

- 1 WHAT?**
What does this test actually measure?
- 2 WHERE?**
Which organ, cell type or system does it represent?
- 3 WHY?**
Why could it be higher or lower than expected?
- 4 WITH WHAT?**
Which other results should be read alongside it?
- 5 WHAT NEXT?**
Does it call for correlation, a repeat test, another investigation or clinical review?



A laboratory report is a collection of biological measurements — not a diagnosis.

Treat the person, not an isolated number.

Don't just check whether the number is high or low.

Learn what the report is actually telling you.

Lab Reports Decoded takes you beyond isolated reference ranges and shows how common laboratory results connect — so a page of numbers becomes a story you can follow.

● Blood & CBC interpretation

● Kidney & liver panels

● Lipids & cardiac markers

● Urine & stool reports

● Fertility & pregnancy tests

● Pattern-based interpretation

● Diabetes & metabolic testing

● Thyroid & hormones

● Vitamins & electrolytes

● Autoimmune & immunology

● Coagulation & ABG

● 10 practical sample reports

FROM NUMBERS → TO PATTERNS → TO UNDERSTANDING

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